

ALL PROGRAMMABLE

ANY MEDIA

5G

4K/8K

ANY STANDARD

ANY MACHINE

ANY NETWORK

5G Wireless • Embedded Vision • Industrial IoT • Cloud Computing



PYNQ™  First steps with PYNQ-Z2

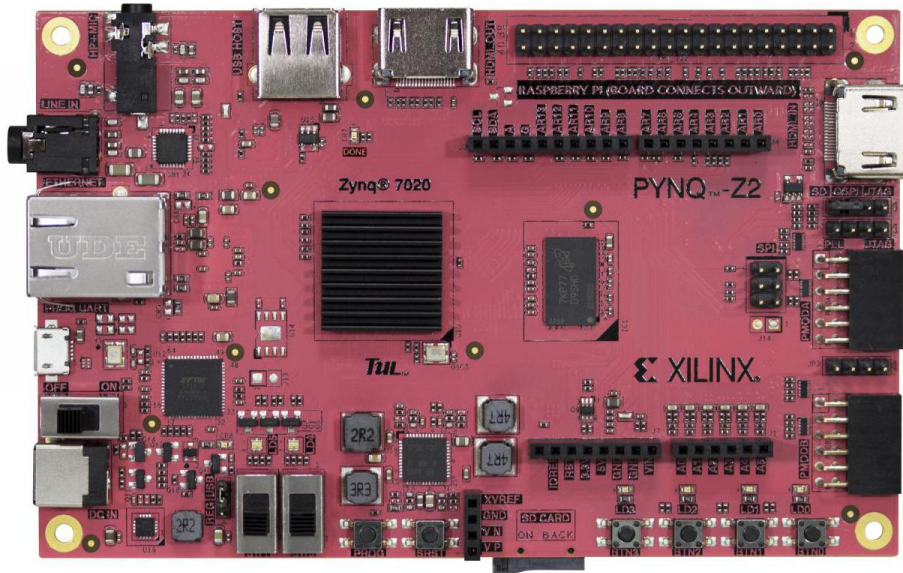


Agenda

- Configure board, SD card
- Connect to the board
- Login in to portal
- Jupyter Notebook
- Labs
 - Getting started with Notebooks
 - Getting started with IPython
 - Exploring PYNQ-Z2
 - Programming on-board peripherals

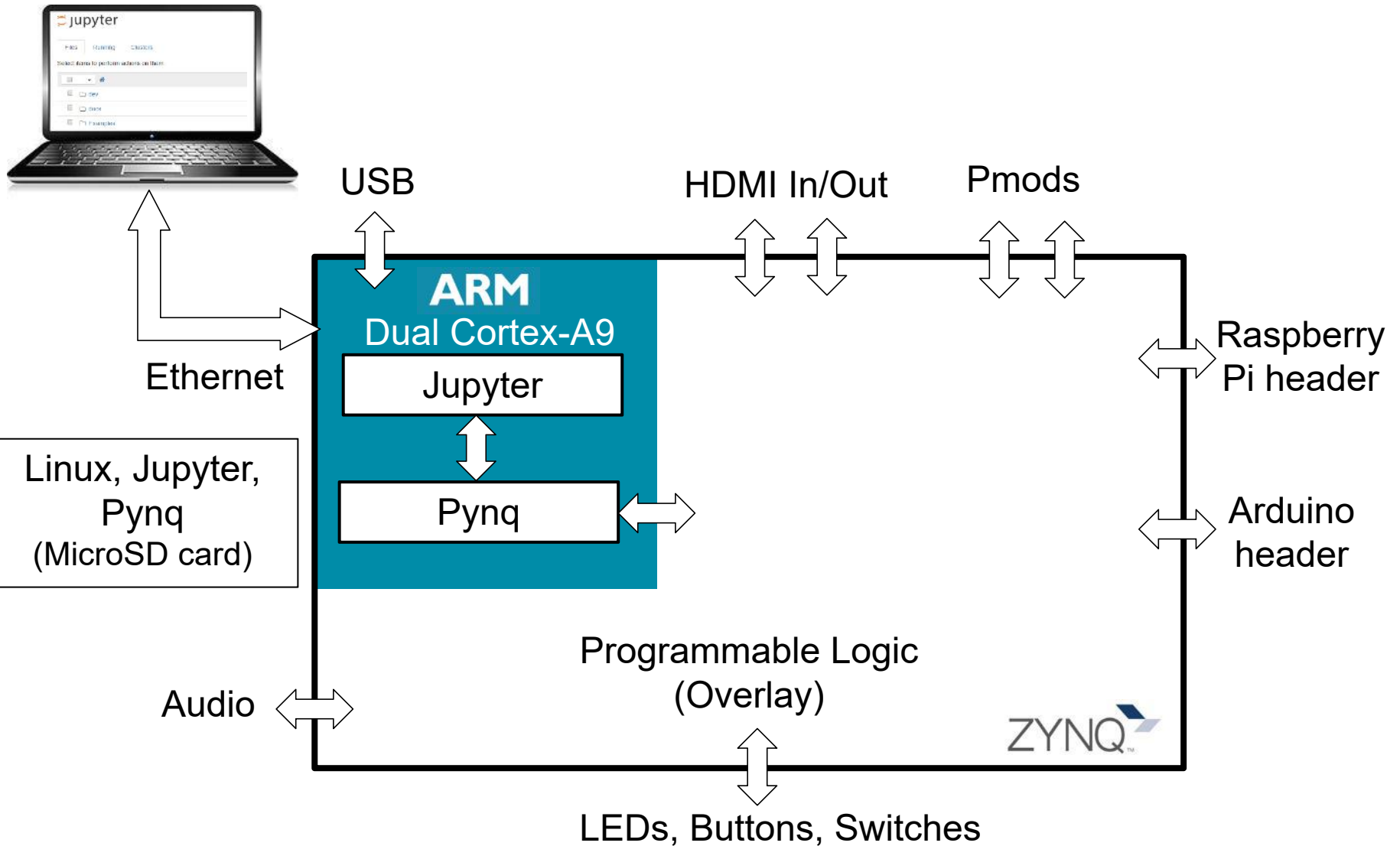
Prerequisites

- PYNQ-Z2 board
- SD Card with preloaded Image*
- Ethernet Cable
- USB cable and available USB port



*Instructions to download and prepare SD card:
http://www.tul.com.tw/download/pynq_z2_v2.2.img.zip

Pynq overview



Connecting to the board

1. Configure board to boot from SD Card
2. Insert SD Card
3. Connect USB cable
4. Connect Ethernet cable to PC or to a Switch/Router
5. Power On

Connecting to the board – Direct connection

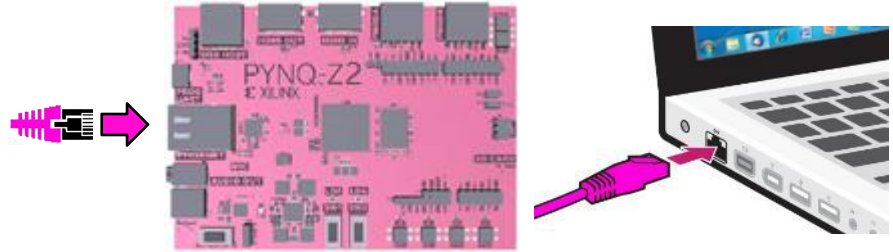
➤ Connect board directly to Ethernet port on PC

- USB to Ethernet adapter if no Ethernet port available

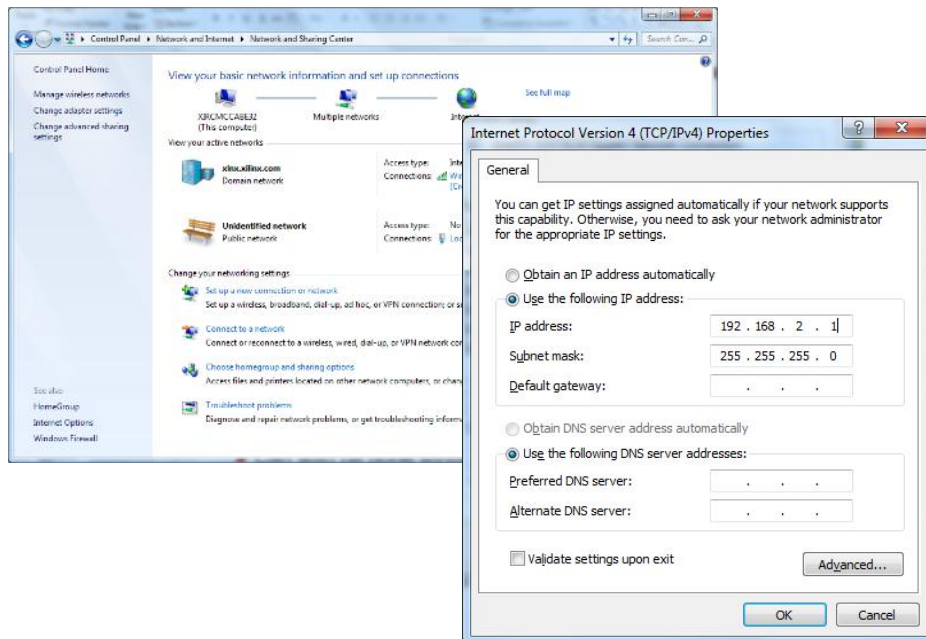
➤ Board IP will default to 192.168.2.99

➤ Manually specify static IP for PC

- Must be in same range as board:
 - E.g. 192.168.2.1
- No internet access unless you bridge network connections (packages can't be updated)

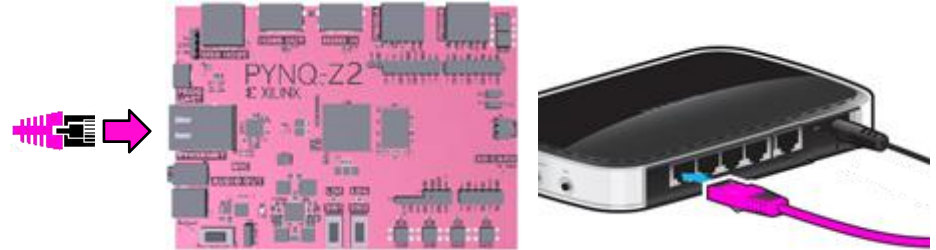


Connect board directly to PC

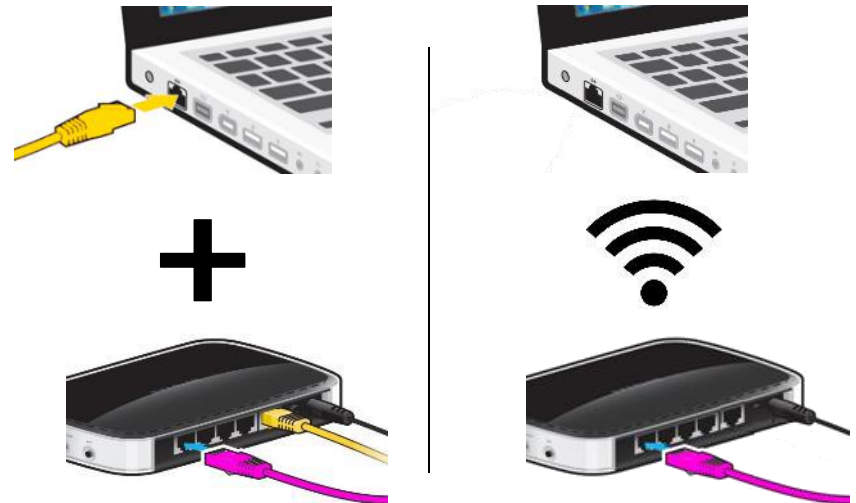


Connecting to the board – Via network switch

- Board is connected to network (switch/router)
 - If DHCP server running on network board will get IP from DHCP server
- Internet access via network allows Python packages to be installed/updated
- PC can connect to same network
 - Wired
 - WiFi



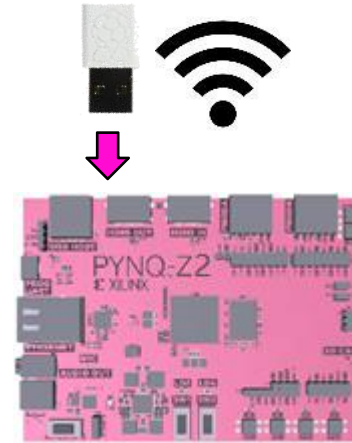
① Connect board to switch/router



② PC connected to router via cable or wireless

Connecting to the board – Via WiFi

- Connect USB WiFi dongle to board
- Connect board and PC to wireless hotspot
 - Router, Mobile Phone
- Need to configure board to connect to WiFi network
 - Notebook available to do configuration
 - `usb_wifi.ipynb`
 - Manually configure from terminal
 - `Modify /etc/network/interfaces.d/`
 - `iwconfig` in next release



① Connect WiFi dongle to board

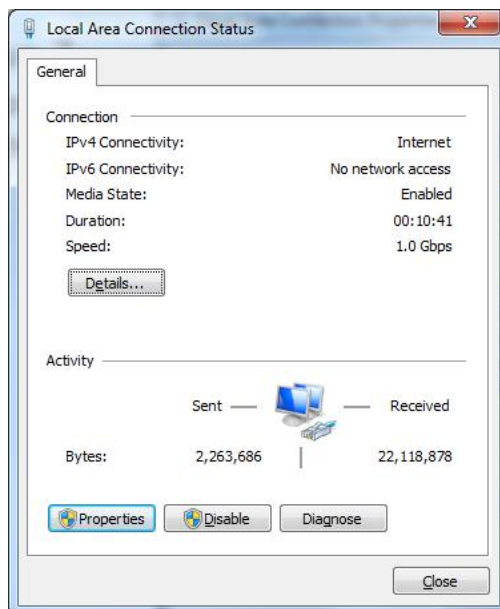
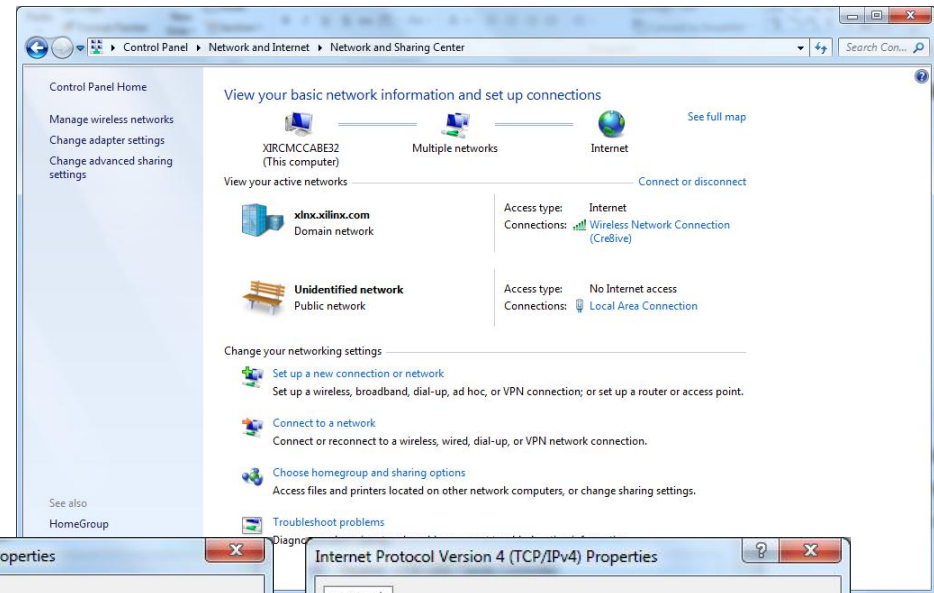
② Connect board and PC to wireless hotspot



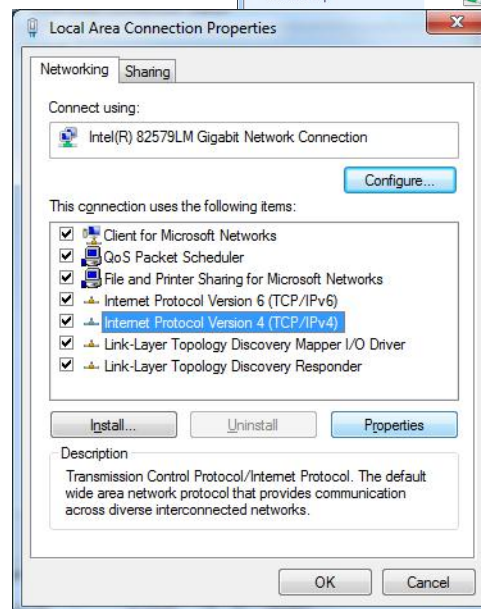
Configure Ethernet - Windows

➤ Windows 7:

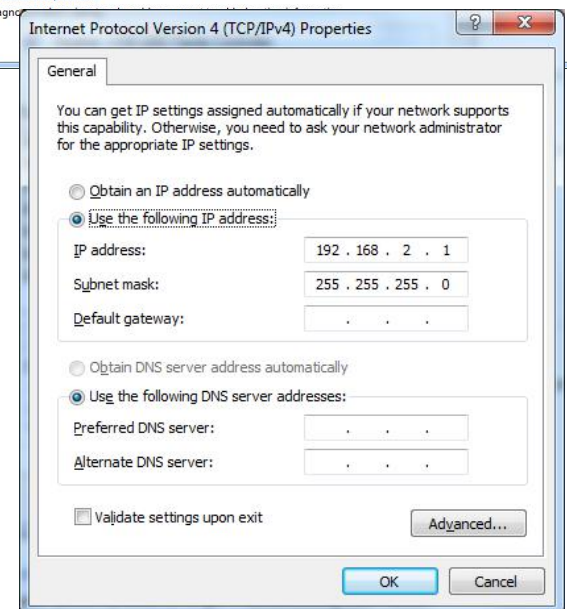
- Control Panel\Network and Internet\Network and Sharing Center



Properties



TCP/IPv4→Properties

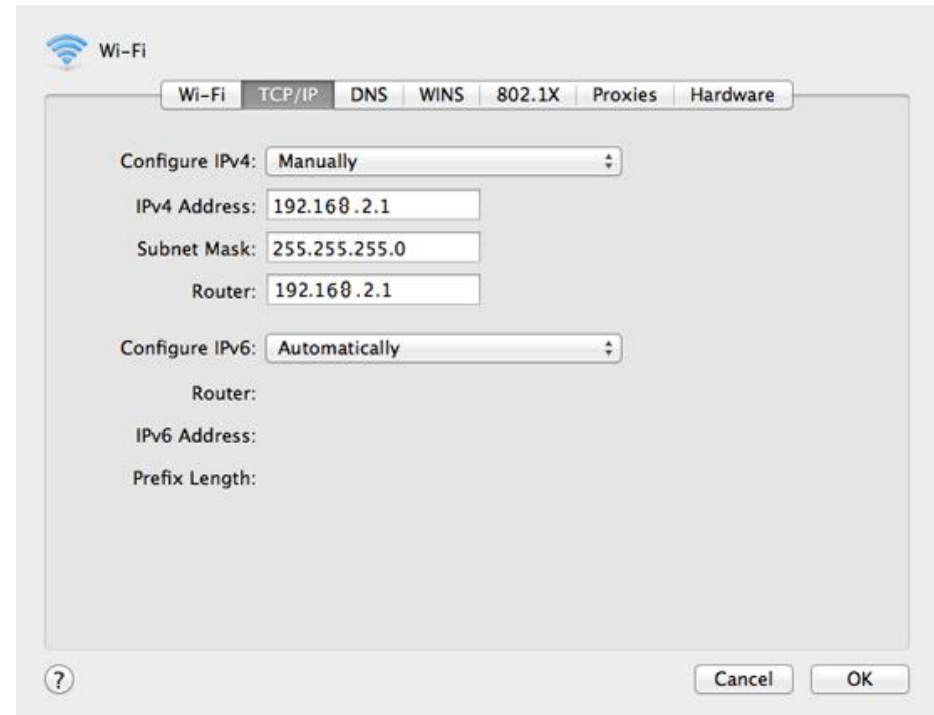


Set IP address

Configure Ethernet - Mac

➤ Mac OS:

- **Apple menu > System Preferences > Network**
- **Select interface > Advanced**



Troubleshooting: LEDs, serial connection

➤ Status LEDs

- Power On: Red LED
- Bitstream Loaded (Pynq booting): Green “Done” LED

➤ USB cable provides serial connection to the board

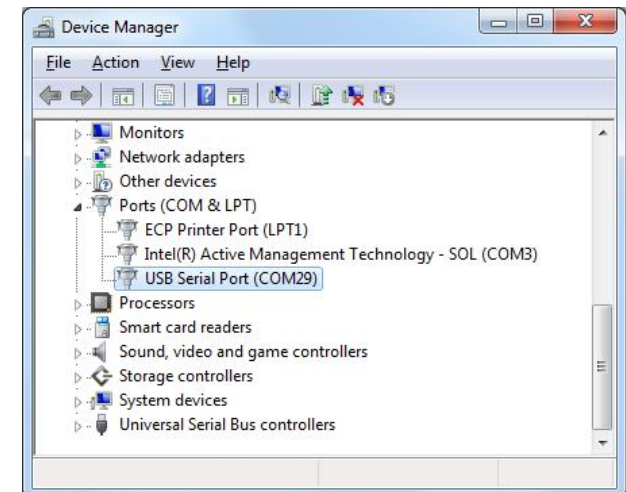
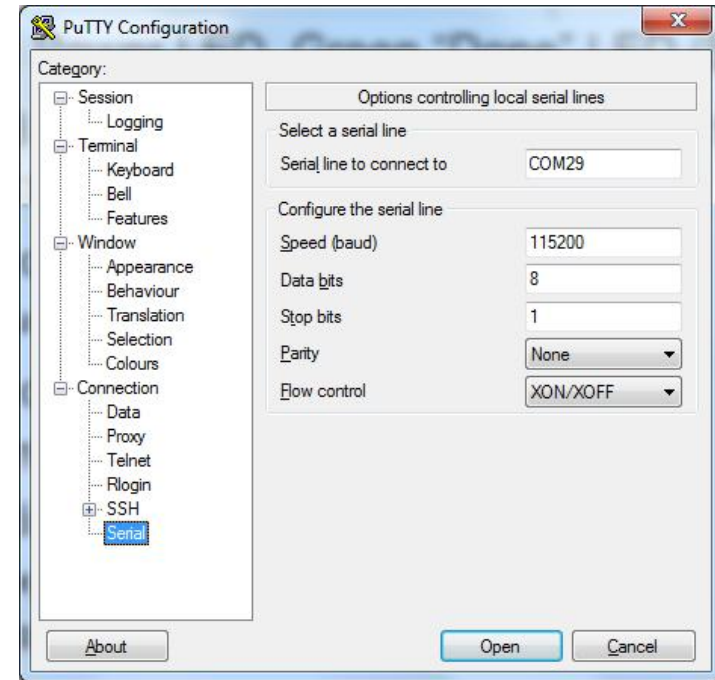
- 115200 baud (data:8 bit, stop bits:1)
- Windows: Device manager to get COM port
- Check board has booted
 - Is Linux shell available on terminal?

➤ Check/Modify board IP

- ifconfig to check network settings

➤ Modify any other Linux settings

- Proxy, Network settings
- sudo password: xilinx



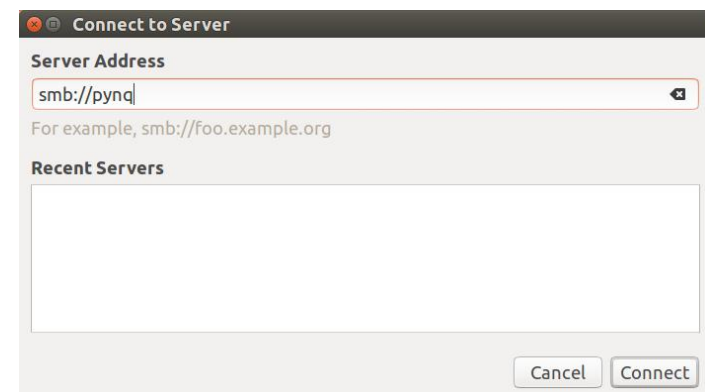
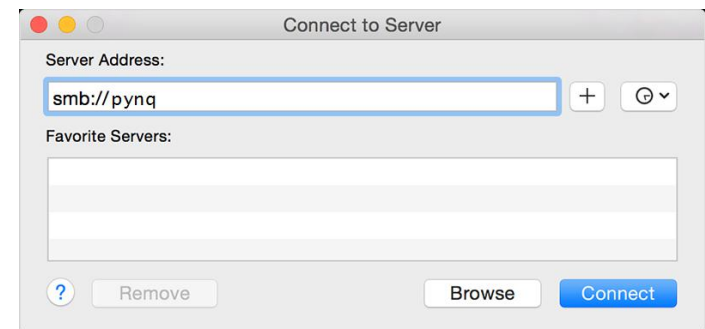
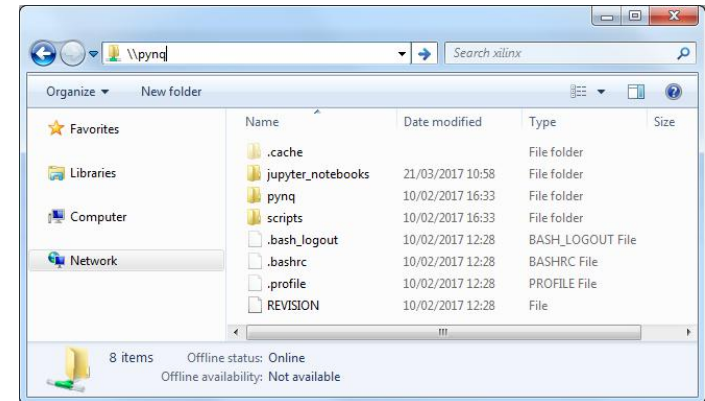
Hostname

- Linux is running on the board
- Change the hostname if multiple boards are on the same network
 - Default hostname is **pynq**
 - Linux username:password = xilinx:xilinx

```
sudo /home/xilinx/scripts/hostname.sh A_UNIQUE_HOSTNAME
```

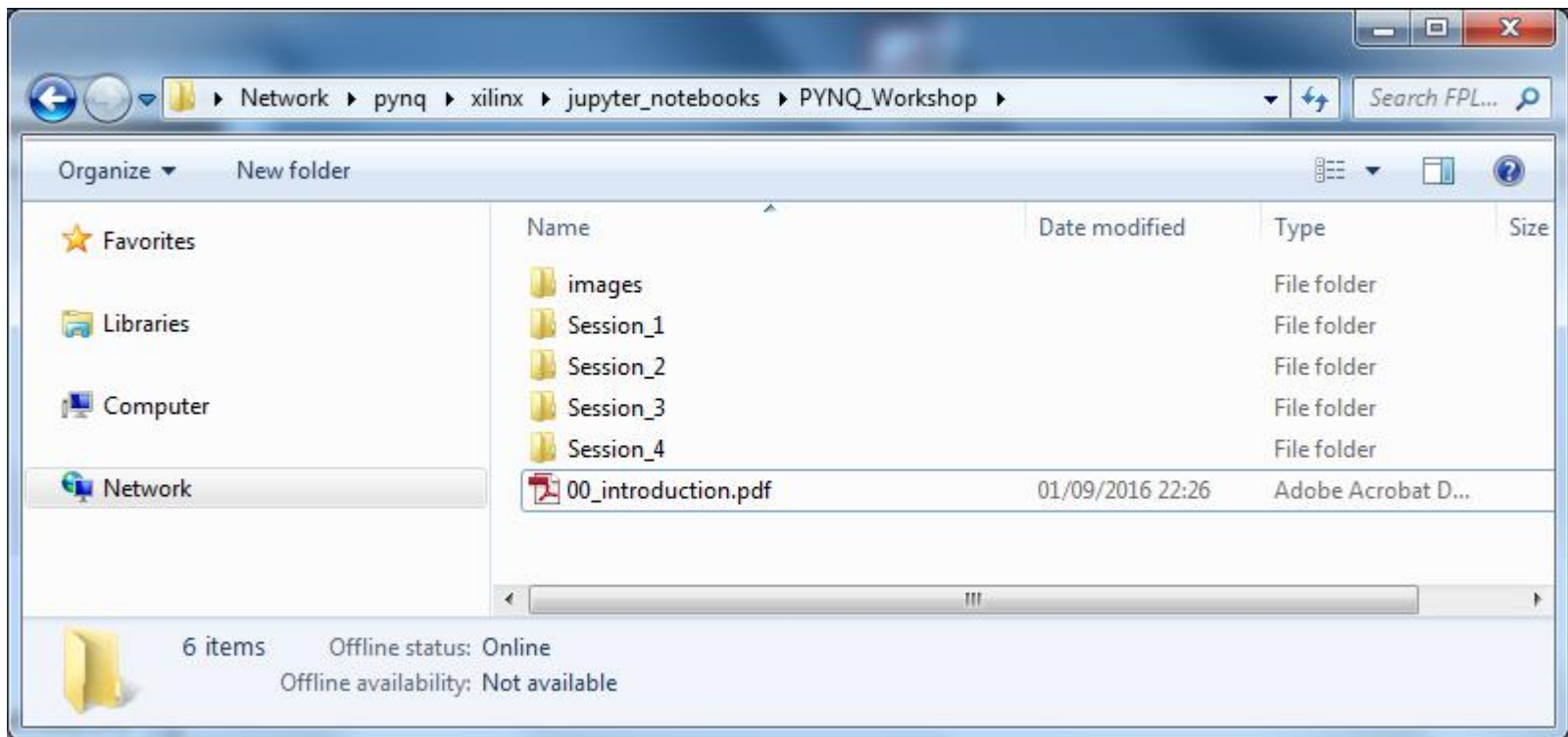
Samba share

- Board can also be accessed as a shared drive
 - Windows: \\pynq\xilinx
 - Linux: smb://pynq/xilinx
 - username:password = xilinx:xilinx
- Copy files between PC and board
- Access documentation/scripts/Notebooks



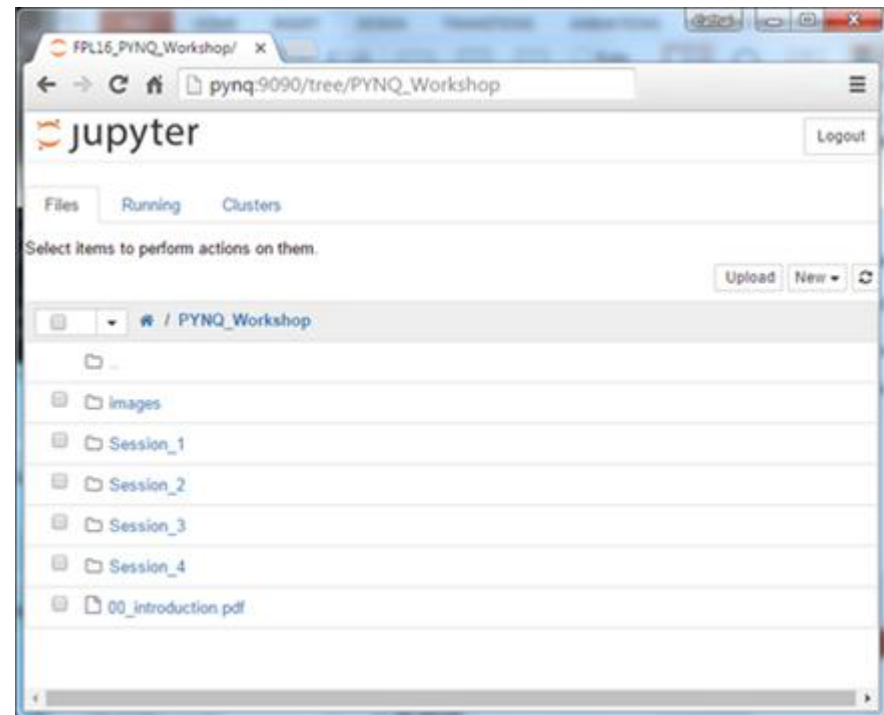
Introduction to Jupyter notebooks

- Copy PYNQ_Workshop folder from USB to:
\\pynq\xilinx\jupyter_notebooks



Log in to portal

- Open a browser*
 - Chrome, Safari, Firefox
- Browse to: `http://pynq:9090` and log in
 - Where *pynq* is the hostname of the board
 - Use the IP address instead of the hostname
- username:password =
xilinx:xilinx
- Browse to PYNQ_Workshop folder



* <http://jupyter-notebook.readthedocs.io/en/latest/notebook.html#browser-compatibility>

Lab exercises: Session 1

- 1 Getting started with Jupyter Notebooks
- 2 Getting started with IPython
- 3 Exploring PYNQ-Z2
- 4 Programming on-board peripherals

Lab Review

➤ Jupyter Notebook

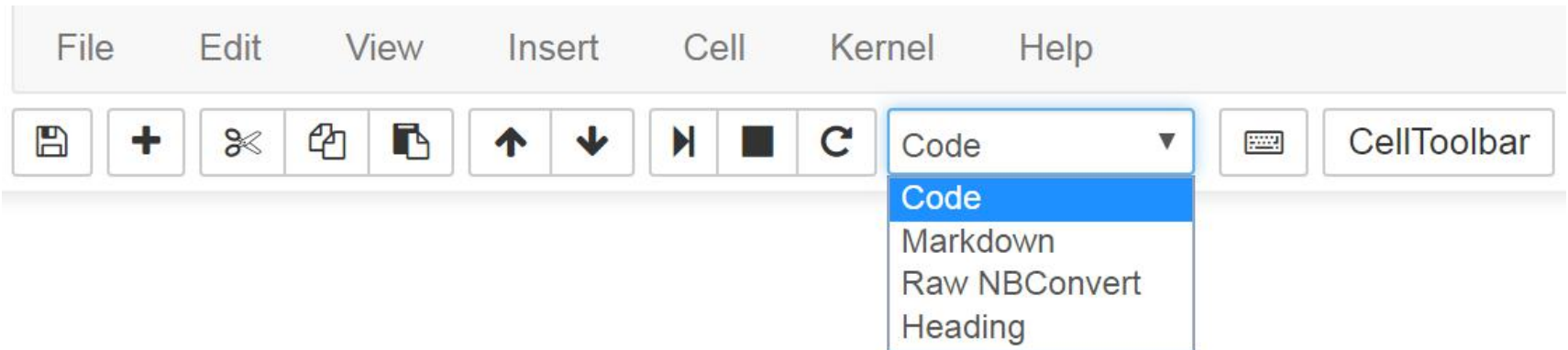
- Web application/server
- Create and share documents
 - Live code, equations, visualizations, explanatory text/comments, results

➤ Make a copy of notebook before editing

➤ Sharing Notebooks

- Save the iPython Notebook .ipynb
- Save as... HTML to share results

Notebook Cells



- Code
 - Python
- Markdown
 - Comments and notes
- Execute cells to run code/render
- Split and reorder cells
- Output printed after cell

```
In [1]: import time
        for i in range (1,9):
            print(i*i)
            time.sleep(0.5)
```

```
1
4
9
16
25
36
49
64
```

Markdown cells

➤ # Headings

➤ * *Italics* *

➤ ** **bold** **

- * Bullet points

- * Bullet points

➤ <html></html>

➤ Not just comments. Explanatory text, notes, documentation.

Notes, comments and Markdown

You can write notes and comments using the `<a href="https://en.wikipedia.org/wiki/Markdown">Markdown Language</a>`.

This cell is a Markdown cell. Double click it now to see the raw markdown.

This is **bold**, this is *italic* and you can see how headings and sub-headings are indicated above using "#". You can also use `<b>html</b>` in markdown.

Execute the cell to render the markdown.

* Note in the dropdown box in the toolbar that this is a Markdown cell.

Shell commands, Cell magics

- Execute shell commands directly from notebook
- Prefix !

```
!uname -a
!whoami
!pwd
!ls
!ping www.xilinx.com
```

- Built in Cell magics

```
%lsmagic
```

Available line magics:

```
%alias %alias_magic %autocall %automagic %autosave %bookmark %cat %cd %
clear %colors %config %connect_info %cp %debug %dhist %dirs %doctest_mo
de %ed %edit %env %gui %hist %history %killbgscripts %ldir %less %lf
%lk %ll %load %load_ext %loadpy %logoff %logon %logstart %logstate %l
ogstop %ls %lsmagic %lx %macro %magic %man %matplotlib %mkdir %more %
mv %notebook %page %pastebin %pdb %pdef %pdoc %pfile %pinfo %pinfo2 %
popd %pprint %precision %profile %prun %psearch %psource %pushd %pwd %
pycat %pylab %qtconsole %quickref %recall %rehashx %reload_ext %rep %re
run %reset %reset_selective %rm %rmdir %run %save %sc %set_env %store
%sx %system %tb %time %timeit %unalias %unload_ext %who %who_ls %whos
%xdel %xmode
```

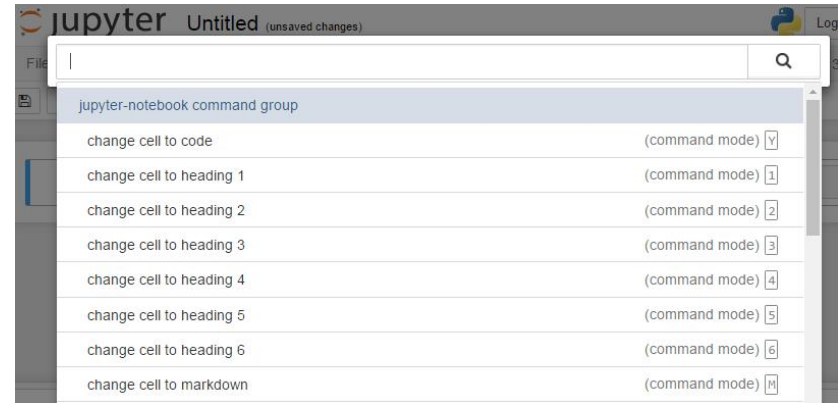
Available cell magics:

```
%%! %%HTML %%SVG %%bash %%capture %%debug %%file %%html %%javascript %
%js %%latex %%perl %%prun %%pypy %%python %%python2 %%python3 %%ruby %
%script %%sh %%svg %%sx %%system %%time %%timeit %%writefile
```

Automagic is ON, % prefix IS NOT needed for line magics.

Help

- Command palette:
Ctrl + Shift + p keys
- Python help:
 - Execute “?” In a cell
 - Help()
- %magic – learn about Ipython magics
 - %lsmagic – get list of magics
 - %quickref



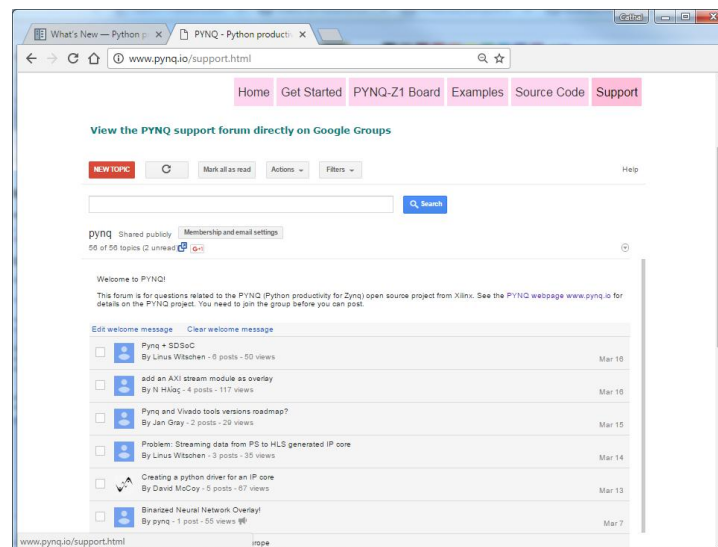
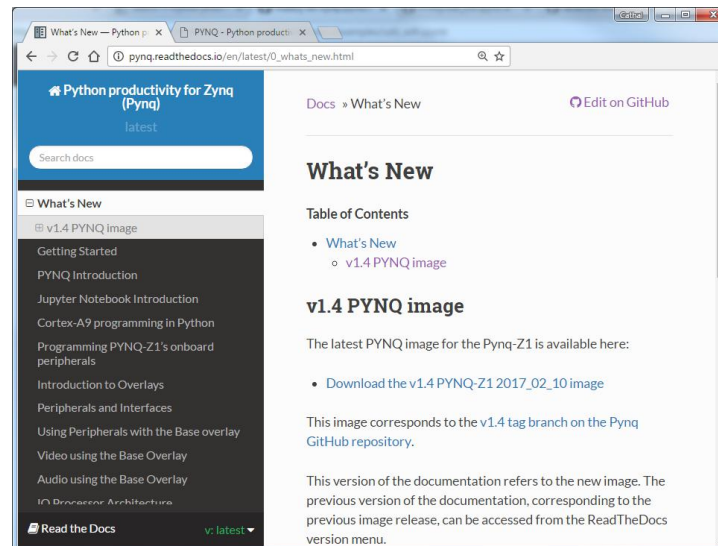
```
In [6]: %lsmagic
Out[6]: Available line magics:
%alias %alias_magic %autocall %automagic %autosave %bookmark %cat %cd %
clear %colors %config %connect_info %cp %debug %dhist %dirs %doctest_mo
de %ed %edit %env %gui %hist %history %killbgscripts %ldir %less %if
%lk %ll %load %load_ext %loadpy %logoff %logon %logstart %logstate %l
ogstop %ls %lsmagic %lx %macro %magic %man %matplotlib %mkdir %more %
mv %notebook %page %pastebin %pdb %pdef %pdoc %pfile %pinfo %pinfo2 %
popd %pprint %precision %profile %prun %psearch %psource %pushd %pwd %
pycat %pylab %qtconsole %quickref %recall %rehashx %reload_ext %rep %re
run %reset %reset_selective %rm %rmdir %run %save %sc %set_env %store
%sx %system %tb %time %timeit %unalias %unload_ext %who %who_ls %whos
%xdel %xmode

Available cell magics:
%%! %%HTML %%SVG %%bash %%capture %%debug %%file %%html %%javascript %
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%script %%sh %%svg %%sx %%system %%time %%timeit %%writefile

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```

Documentation and Support

- Documentation
 - <http://pynq.readthedocs.io>
- Support
 - <http://www.pynq.io/support>
- GitHub
 - Issue tracker



Next steps

➤ examples/

- Peripherals, Grove, Pmod
- How to use overlays
- Other peripherals, PS/PL



Files

Running

IPython Clusters

Select items to perform actions on them.

⌂ / Examples

	..
	data
	audiovideo_sobel.ipynb
	board_ps_clocks.ipynb
	board_sws_btns_leds.ipynb
	overlay_download.ipynb
	overlay_integration.ipynb
	pmod_adc.ipynb
	pmod_als.ipynb

Summary

- Configure board, SD Card
- Connect to the board
- Log In to portal
- Jupyter Notebooks
- Labs
- Help

Questions?